

Double Integrals



Double integrals over a rectangle

- 1 Set up the double integral of $f(x, y) = xy$ over the rectangle $[0, 2] \times [0, 3]$ (integrate y first).
 - 2 Set up the double integral of $f(x, y) = x^2 + y$ over the rectangle $[1, 3] \times [0, 2]$ (integrate x first).
-

Iterated integrals

- 3 Evaluate $\int_0^2 \int_0^3 xy \, dy \, dx$.
 - 4 Evaluate $\int_0^1 \int_0^2 (x^2 + y) \, dy \, dx$.
 - 5 Evaluate $\int_0^\pi \int_0^1 y \sin x \, dy \, dx$.
-

Fubini's Theorem: order of integration

- 6 Verify Fubini's Theorem for $f(x, y) = xy$ on $[0, 1] \times [0, 2]$ by evaluating the integral in both orders.
 - 7 State the hypothesis of Fubini's Theorem needed for the two iterated orders to be guaranteed equal.
-

Double integrals over general regions

- 8 Evaluate $\int_0^1 \int_0^x (x + y) \, dy \, dx$.
 - 9 Set up (do not evaluate) the double integral of $f(x, y) = xy$ over the triangular region bounded by $y = 0$, $x = 1$, and $y = x$.
 - 10 Evaluate $\int_0^1 \int_y^1 2 \, dx \, dy$ (a region bounded by $x = y$, $x = 1$, $y = 0$).
-

Reversing the order of integration

- 11 Reverse the order of integration for $\int_0^1 \int_0^x f(x, y) \, dy \, dx$, describing the new bounds.
- 12 Evaluate $\int_0^1 \int_x^1 e^{y^2} \, dy \, dx$ by first reversing the order of integration (the inner integral has no elementary antiderivative in the original order).

Polar coordinates

- 13 Convert $\iint_D (x^2 + y^2) dA$ to polar coordinates, where D is the disk $x^2 + y^2 \leq 4$, and evaluate.
- 14 Evaluate $\iint_D xy dA$ where D is the quarter disk $x^2 + y^2 \leq 1$, $x \geq 0$, $y \geq 0$, using polar coordinates.
- 15 Use polar coordinates to evaluate $\iint_D e^{-(x^2 + y^2)} dA$ where D is the entire plane $\mathbb{R}^2$ (the classic Gaussian integral setup).
-

Area and average value

- 16 Use a double integral to find the area of the region bounded by $y = x^2$ and $y = x$ (for $0 \leq x \leq 1$, where $x \geq x^2$).
- 17 The average value of $f(x, y)$ over a region D with area A is $\frac{1}{A} \iint_D f dA$. Find the average value of $f(x, y) = xy$ over the rectangle $[0, 2] \times [0, 3]$, using the earlier result $\iint_D xy dA = 9$.
- 18 Use a double integral to find the area of the region bounded above by $y = 4 - x^2$ and below by $y = 0$.
-

Volume via double integrals

- 19 Set up (do not evaluate) the double integral for the volume under the surface $z = x^2 + y^2$ above the rectangle $[0, 1] \times [0, 1]$.
- 20 Evaluate the volume integral set up above.
- 21 Find the volume of the solid bounded above by $z = 6 - x - y$ and below by $z = 0$, over the triangle in the xy -plane with vertices $(0, 0)$, $(6, 0)$, $(0, 6)$ (i.e., $0 \leq x \leq 6$, $0 \leq y \leq 6 - x$).
-

Double integrals in applications: mass and center of mass

- 22 A flat plate occupies the rectangle $[0, 2] \times [0, 1]$ with density $\rho(x, y) = x + y$. Set up the double integral for its total mass $M = \iint_D \rho dA$.
- 23 Evaluate the mass integral set up above.
- 24 The x -coordinate of the center of mass is $\bar{x} = \frac{1}{M} \iint_D x \rho dA$. Set up (do not evaluate) this integral for the plate above ($\rho(x, y) = x + y$ on $[0, 2] \times [0, 1]$, $M = 3$).

25 Evaluate $\bar{X}$ from the integral set up above.